

Microservices JWT

Question 1: Implement JWT Authentication

Code / Steps:

Commands:

```
dotnet new webapi dotnet add package  
Microsoft.AspNetCore.Authentication.JwtBearer dotnet run
```

Sample Terminal Output:

```
Terminal Output:  
Build succeeded.  
Now listening on: https://localhost:5001
```

```
POST /api/auth/login  
Status: 200 OK
```

```
{  
  "token": "eyJhbGciOiJIUzI1NiIs..."  
}
```

Question 2: Secure API Endpoint

Code / Steps:

Request:

```
GET /api/secure/data  
Authorization: Bearer
```

Sample Terminal Output:

```
Terminal Output:  
Status: 200 OK This is  
protected data.
```

```
Without token:  
Status: 401 Unauthorized
```

Question 3: Role-Based Authorization

Code / Steps:

Request:

```
GET /api/admin/dashboard  
Authorization: Bearer
```

Sample Terminal Output:

```
Terminal Output:  
Status: 200 OK Welcome to the  
admin dashboard.
```

```
Non-admin:  
Status: 403 Forbidden
```

Question 4: Expired JWT Handling

Code / Steps:

Request with expired token:
GET /api/secure/data

Sample Terminal Output:

Terminal Output:
Status: 401 Unauthorized
Header:
Token-Expired: true